

23.02.26

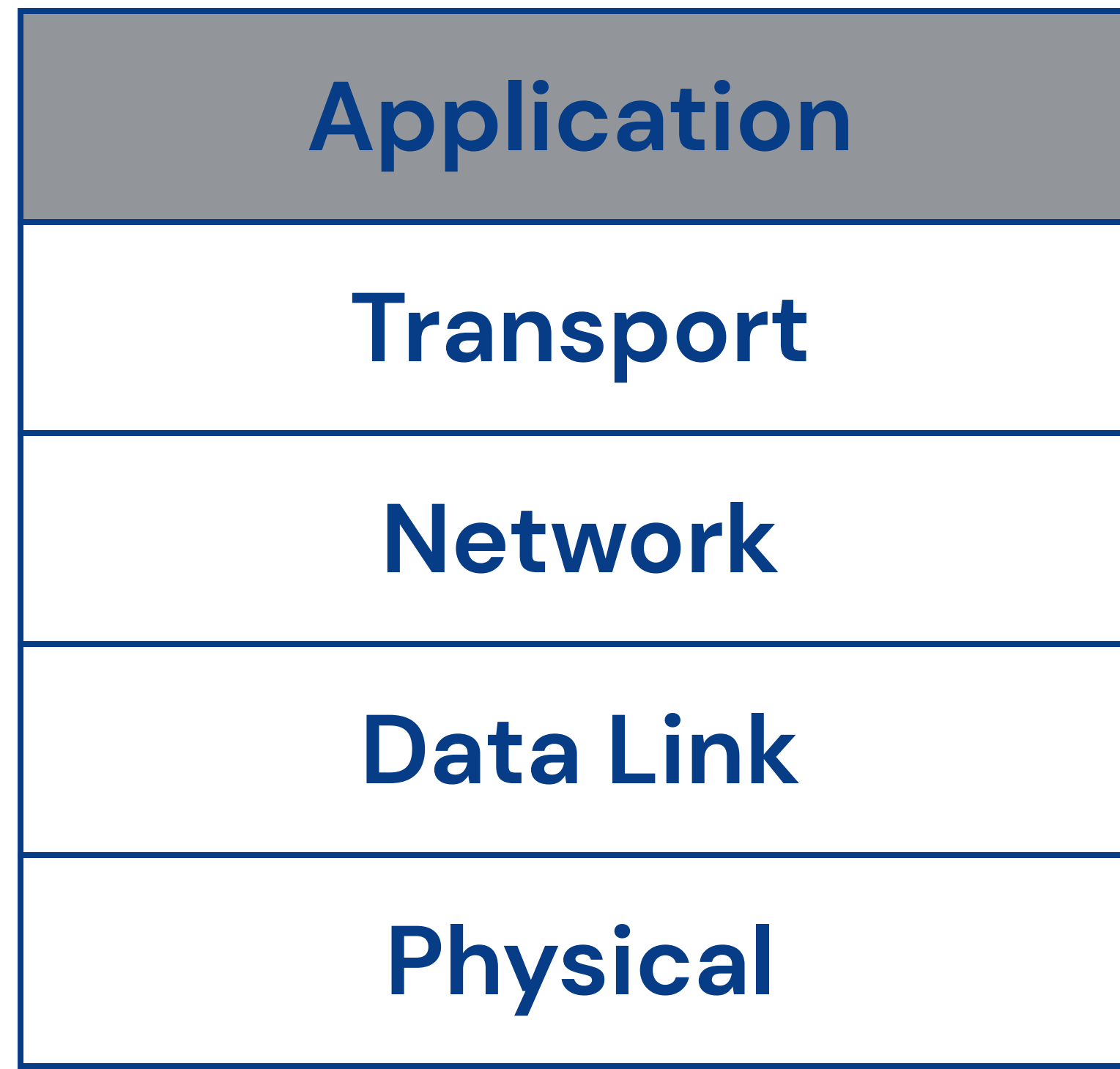
COMPUTER NETWORKS

CSE 321

Application Layer → user (বা) communicate করার interface দেয়

Top layer of the OSI model.

It provides the services and protocols that end-users and their applications interact with directly.



Network Application Architecture

Two predominant architectural paradigms:

1. **Client-Server** → no direct communication b/w clients. server এর সাথে communication.

2. **Peer-to-Peer (P2P)** e.g. web applications

↓
connected clients

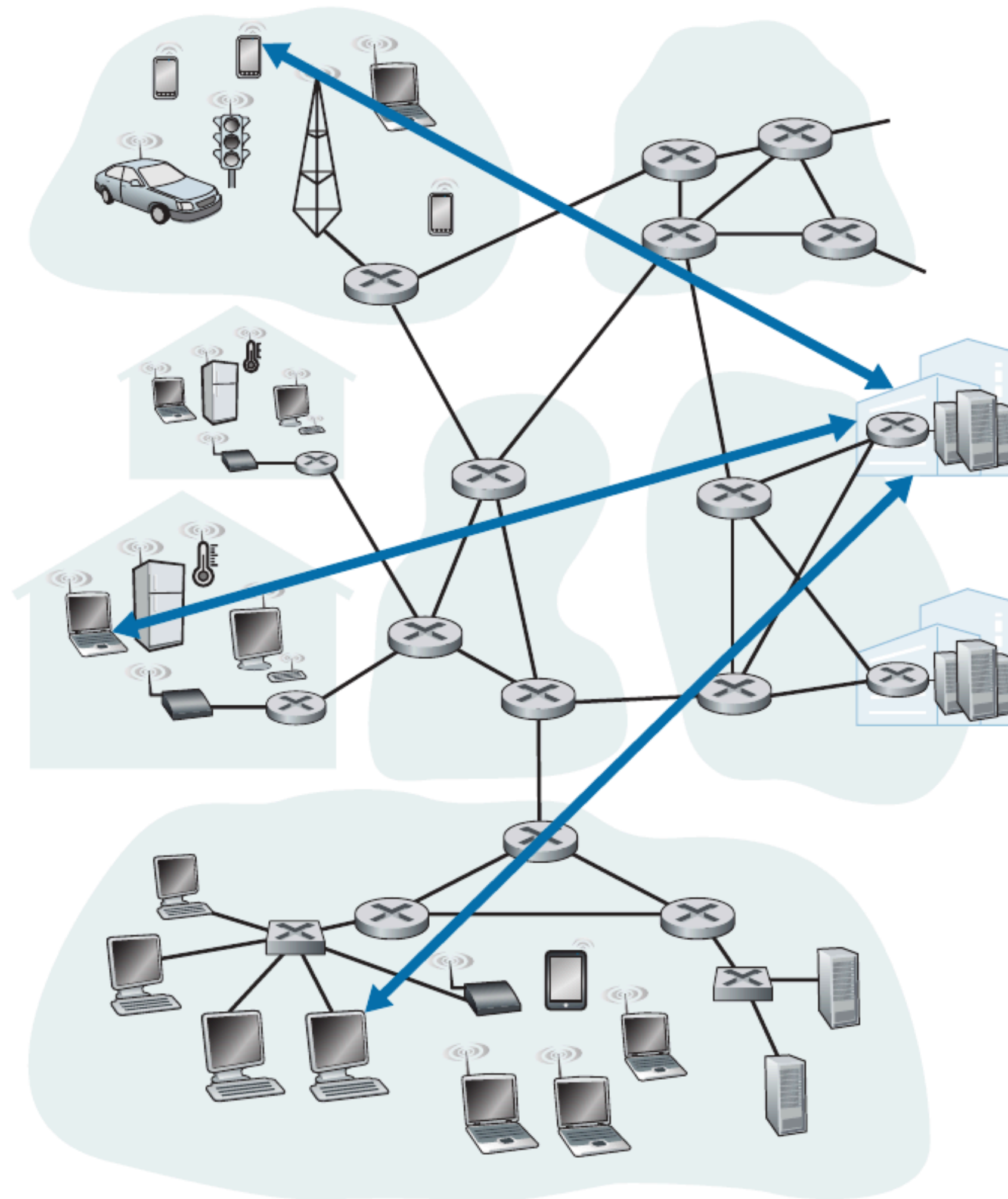
(client বা নিউদের মাধ্যমে communicate করে) no centralized server

e.g. torrent. multiple user এর কাছে file থাকে যারা থেকে আরও বান্ধে আসে,

↳ malicious user থাকতে পারে (malware/virus চলে আসতে পারে)

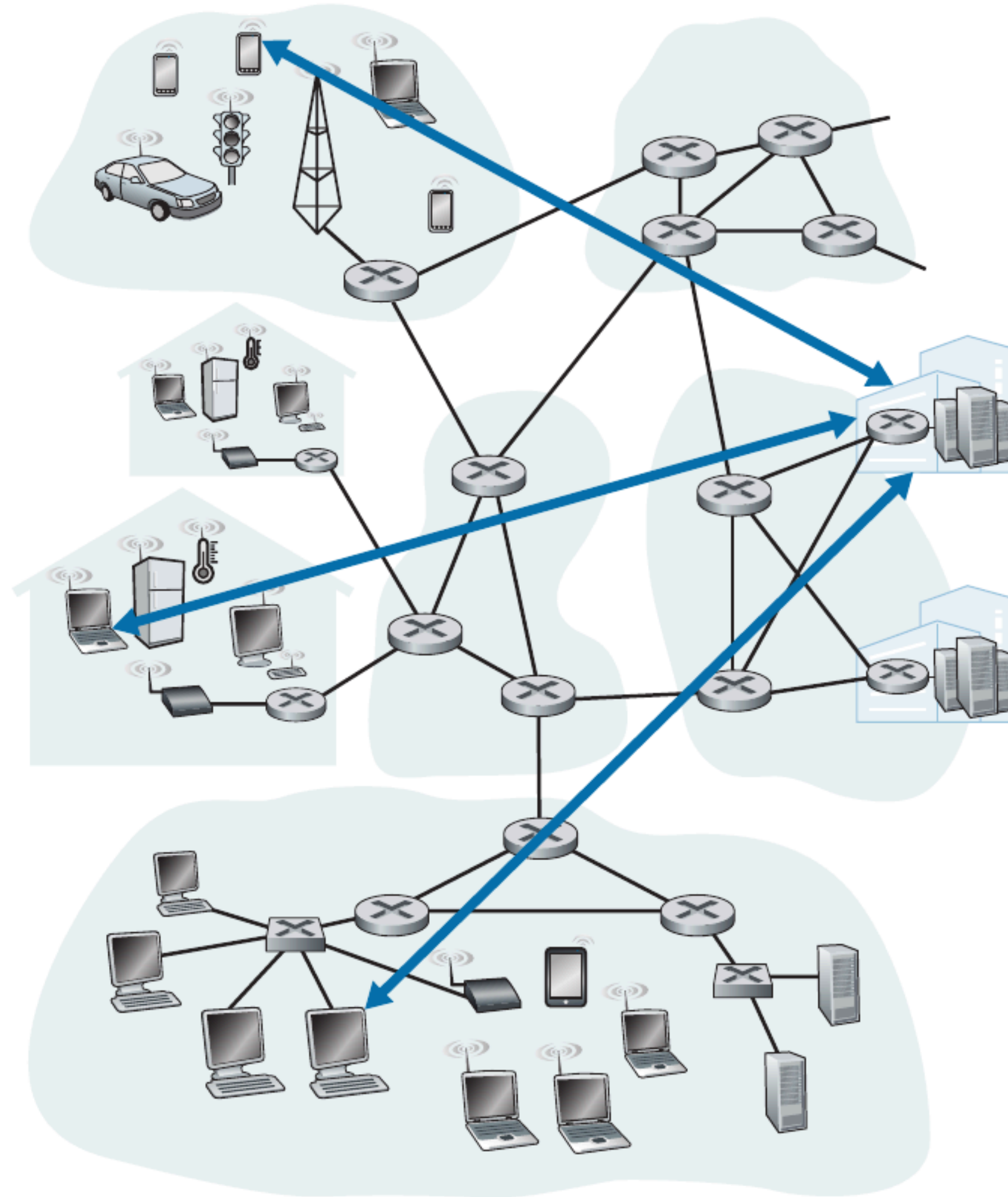
→ illegal (piracy হতে পারে)

Client-Server



Client-Server

**Clients do not
communicate
with each other
directly**

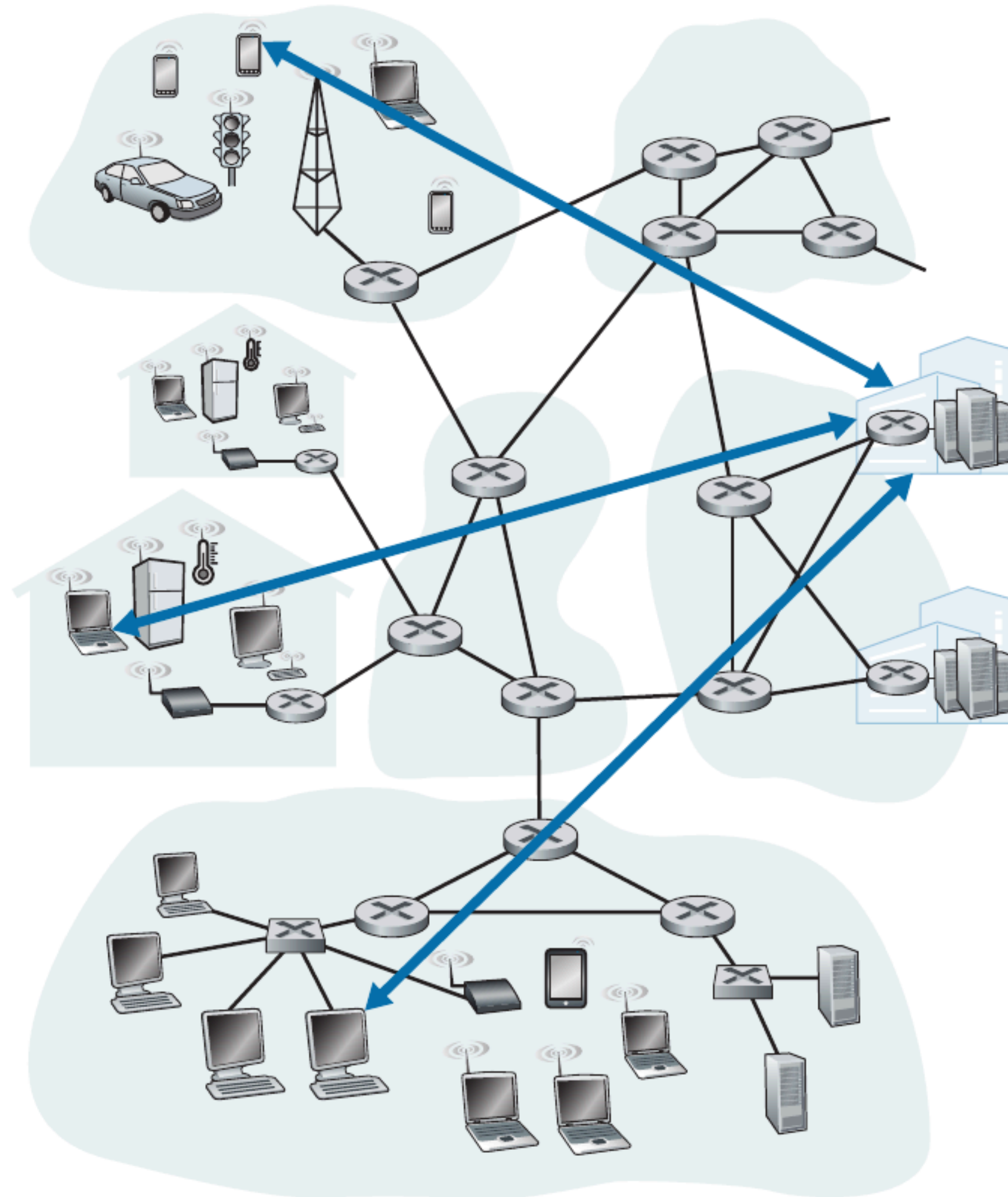


Servers

Client-Server

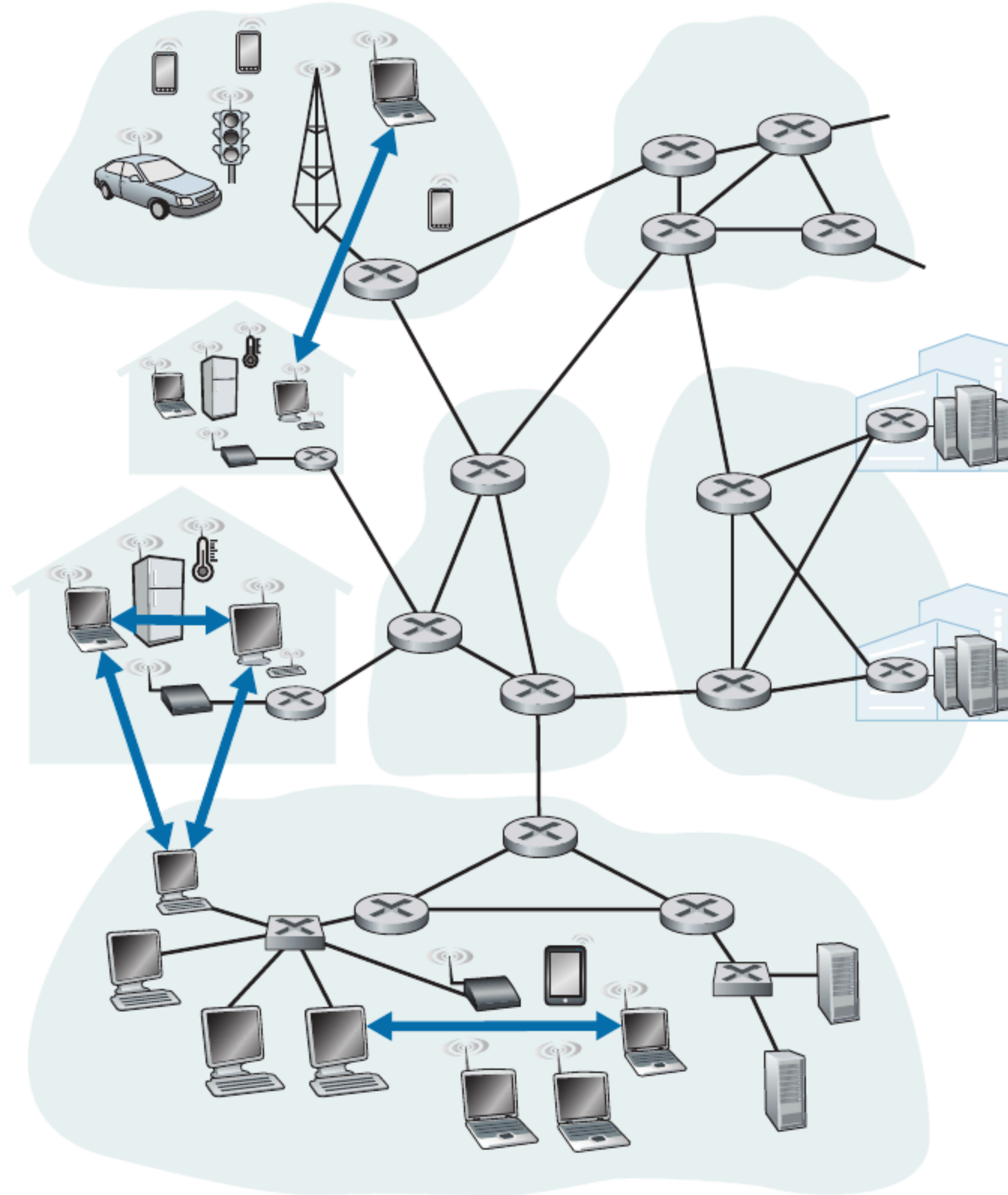
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Example:
Web Applications



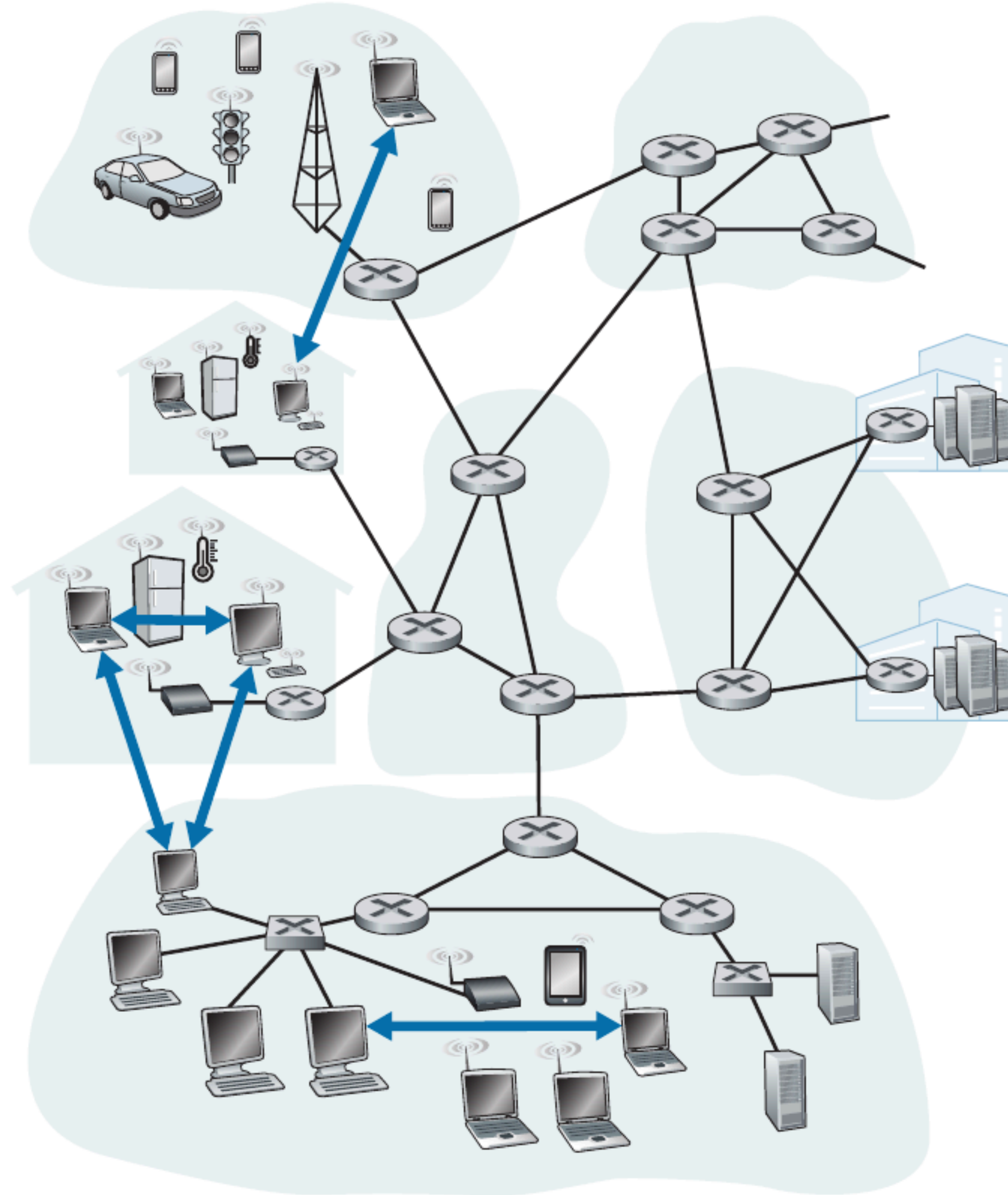
Servers

P2P



P2P

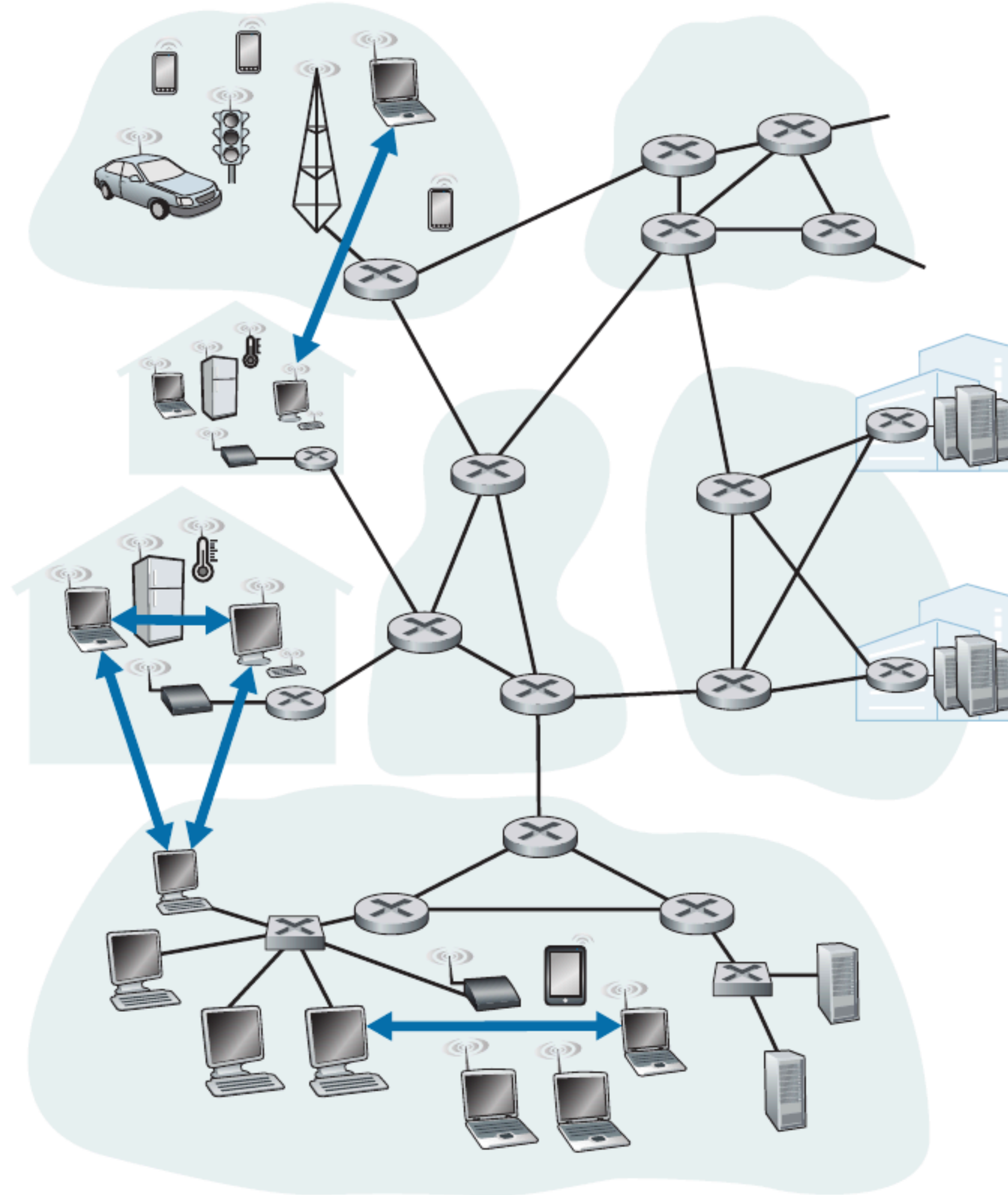
**Direct
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between pairs of
intermittently
connected hosts,
called peers**



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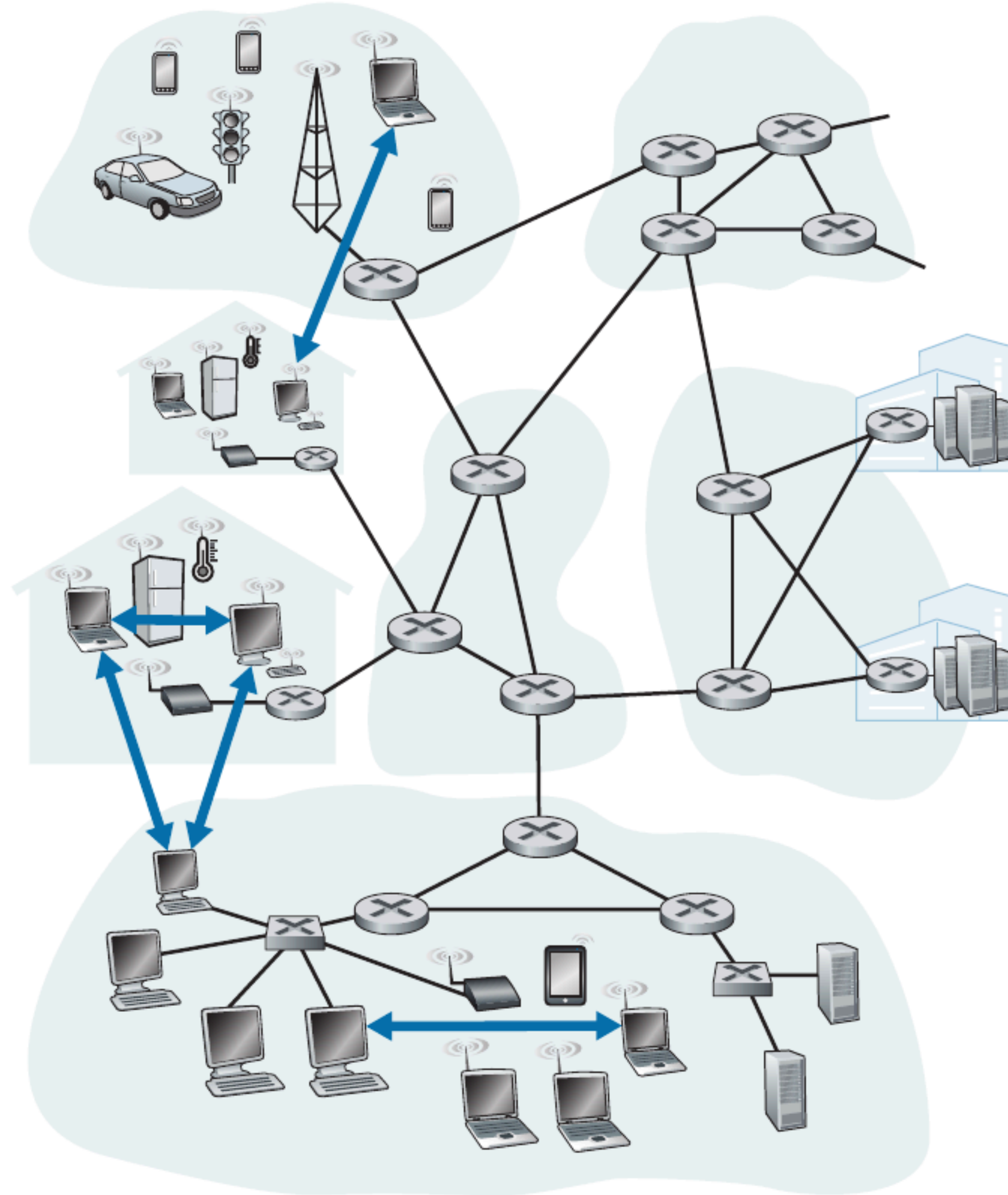
**Example:
Torrenting**



P2P

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Example:
Torrenting



Some Application Layer Protocols

- DHCP
- HTTP
- DNS
- SMTP and IMAP



NETWORK TECHNICIAN

Just a normal day at the job...

mematic.net

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